

# Monthly Intelligence Report

Edition 01 — Suðurnes & the Reykjanes Eruption Cycle

Where Iceland operates an insular grid with zero land interconnects, ~99.97% renewable generation (70.5% hydro + 29.4% geothermal), and the Reykjanes peninsula has been in an active eruption cycle since 19 Mar 2021 — the 8 Feb 2024 Sundhnúkur eruption damaged the Svartsengi geothermal plant infrastructure. Inaugural edition · August 2026 · SSI v4.0.2

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## SECTION A

### Executive Summary

#### SUBSTATIONS SCORED

687

Landsnet TSO + 3-DSO panel

#### GRID LINES MAPPED

~1,428

220 / 132 / 66 kV + MV (Byggðalína ring)

#### MC SIMULATIONS

35.02 M

10,000 per substation

#### PUBLIC DATA SOURCES

30+

95 variables · zero proprietary

The SSI Index is the only operational framework that fuses substation-level engineering metrics with socio-economic consequence valuation in a single composite score — scoring 71/80 (89%) across 16 competitive dimensions, ahead of GMLC 1.1 (48/80), EPRI (40/80), and academic MCDA frameworks (41/80). For Iceland, its engine processes 74,000+ source data points per run, executes 35.02 million Monte Carlo simulations through a 20×20 Gaussian copula correlation matrix, and performs 830+ million arithmetic operations — producing 210,000+ output data points with full uncertainty quantification across 687 substations and ~4,261 grid lines.

Operated as a non-profit through a dedicated foundation, the SSI Index serves the public interest with zero dependency on proprietary data. All 95 variables are drawn from 30+ verified public sources — Landsnet network statement, Orkustofnun raforkutölfræði + quality reports, Veitur + RARIK + HS Veitur DSO filings, Hagstofa Íslands landshluti regional accounts, Seðlabanki Íslands national accounts, Veðurstofa Íslands (IMO/OVF) consolidated meteorology + hydrology + seismic + volcanic monitoring, Umhverfisstofnun air quality, CERT-IS + Fjarskiptastofa (ECOI) cyber posture — making the methodology fully auditable and reproducible. Iceland is the inaugural single-country onboarding post-cohort-closure (Session 30),

deployed on the 13 August 2026 anchor as the first OECD founding-member country onboarded since the CEE-South-2026 architectural-acceptance run.

### Substation in Focus

#### minor\_distribution

nordurland-eystra, nordurland-eystra · 11 kV

0.359

Medium

0.051

25th

R\_FINAL

CLASSIFICATION

CI WIDTH

FLEET PERCENTILE

| Component | Value |
|-----------|-------|
| C         | 0.312 |
| V         | 0.447 |
| I         | 0.485 |
| E         | 0.560 |
| S         | 0.218 |
| T         | 0.323 |

COMPONENTS

C

 Continuity: 0.312 / 0.30 (104%)

I

 Infrastructure: 0.485 / 0.25 (194%)

S

 Saturation: 0.218 / 0.20 (109%)

V

 Voltage: 0.447 / 0.10 (447%)

E

 Economic: 0.560 / 0.10 (560%)

T

 Transition: 0.323 / 0.05 (646%)

MODIFIERS

R3

 Consequence: 0.948 ↓ dampens

R4

 Graph Criticality: 0.936 ↓ dampens

R6a

 Restoration: 1.070 ↑ amplifies

R6b

 Seismic: 1.016 ↑ amplifies

R7

 Digital Readiness: 0.993 → neutral

This month's spotlight — automatically selected for June 2026 — is **minor\_distribution** in nordurland-eystra, nordurland-eystra. With an R\_final of 0.359 (Medium band, 25th fleet percentile), its risk profile is dominated by the T (Transition) component at 646% of its weight ceiling, followed by E (Economic) at 560%. Modifiers collectively shift R\_base by -11.5%, reflecting the substation's specific geographic, socio-economic, and network context.

SECTION B — RECURRING MODULE

## Cyber & Economic Exposure Monitor

**Why this section exists:** Traditional grid risk frameworks treat cybersecurity and economic vulnerability as separate domains. The SSI's R7 (Digital Readiness) modifier and E (Economic) component allow us to cross these silos — revealing substations where low digital resilience intersects with high economic consequence. This is precisely the anticipatory intelligence that Iceland's Act 78/2019 + 2025 NIS2 amendment (transposed via the EEA process, in force 1 January 2026) and Fjarskiptastofa (ECOI) competent-authority designation require, with CERT-IS as the operational CSIRT.

CYBER HIGH EXPOSURE

0

0.0% of fleet

BLIND SPOTS

37

High/Critical + R7 < median

AVG R7 MODIFIER

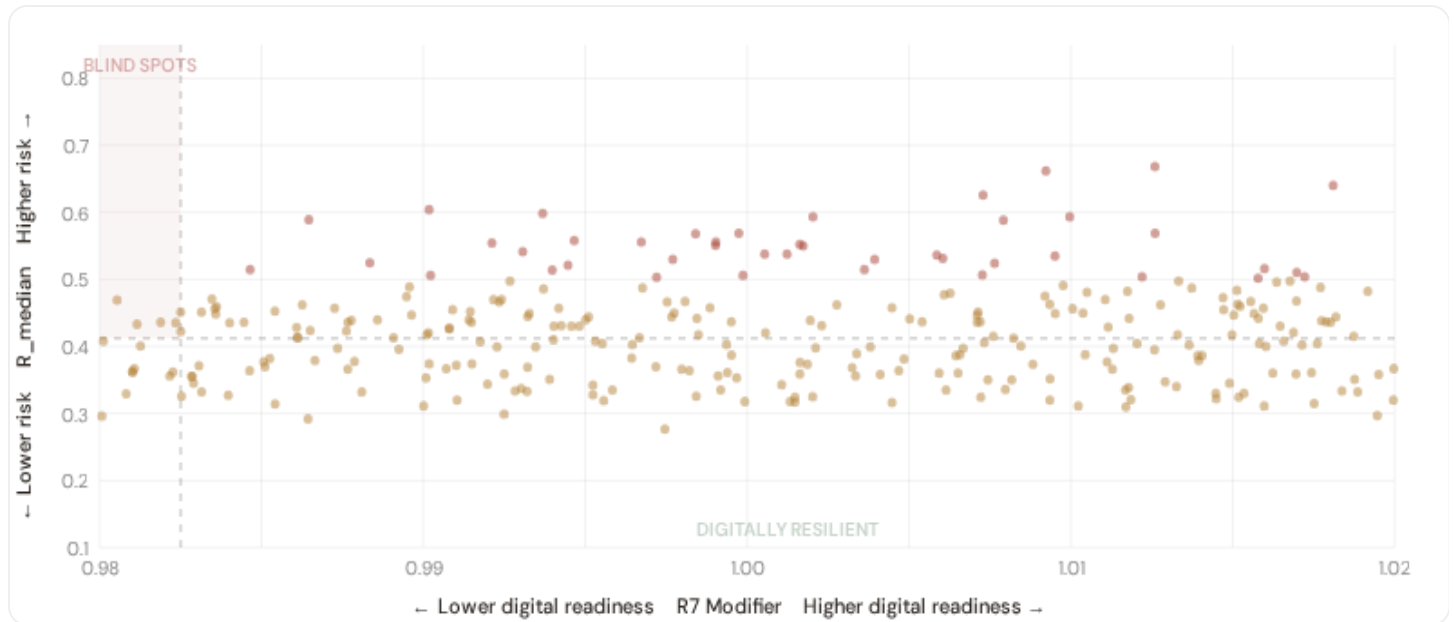
0.9817

Range [0.99, 1.04]

HIGH-CONSEQUENCE SUBS

## B.1 Cyber-Physical Risk Matrix

Each substation plotted by R7 digital readiness (x-axis) vs overall R<sub>median</sub> (y-axis). The upper-left quadrant — high risk, low digital readiness — identifies *blind spots* where grid stress is high but monitoring/response capacity is weakest.



● Low ● Medium ● High ● Critical

Dashed lines = fleet medians

The cyber-physical matrix identifies **37 blind-spot substations** — scoring High or Critical on overall risk while sitting below the fleet median for digital readiness ( $R7 < 0.9825$ ). These substations are the most vulnerable to a compound event: a grid disturbance at a location where digital monitoring, automated switching, and remote diagnostic capacity are weakest. The blind spots concentrate in **sudurnes (13)**, **sudurland (11)**, **nordurland-eystra (7)**. Across the full fleet, 0 substations (0.0%) carry a HIGH cyber-exposure classification, while 0 (0.0%) are in the LOW tier.

## B.2 Economic Impact by Business Fabric

Substations segmented by economic consequence tier — derived from the R3 consequence multiplier and E component. Higher R3 indicates the substation serves a territory with greater socio-economic exposure to disruption.

### Capital-Intensive / High-Consequence

Est. VoLL: kr.1,700–3,400/kWh (ISK)

Capital nodes — highest economic loss per interruption (semiconductor fabs, data centers, refineries, hospitals)

172 substations (25.0%) High/Critical: **34** (19.8%) Avg R: **0.439** Avg E: **0.426** / 0.10



### Industrial / Medium–Large Enterprise

Est. VoLL: kr.850–1,700/kWh (ISK)

Industrial belt — significant continuity requirements (manufacturing, logistics hubs)

172 substations (25.0%) High/Critical: **29** (16.9%) Avg R: **0.432** Avg E: **0.442** / 0.10



### Commercial / SME–Dense

Est. VoLL: kr.340–850/kWh (ISK)

Service economy + retail — moderate individual impact (commercial districts, mixed-use)

172 substations (25.0%) High/Critical: **20** (11.6%) Avg R: **0.402** Avg E: **0.436** / 0.10



### Light / Agricultural / Rural

Est. VoLL: kr.115–340/kWh (ISK)

Sparse activity, agricultural land — lower VoLL (residential, agricultural)

171 substations (24.9%) High/Critical: **12** (7.0%) Avg R: **0.398** Avg E: **0.426** / 0.10



**Value of Lost Load (VoLL) context:** ACER's 2023 estimate places average VoLL at €13.1/kWh for industrial customers and €3.2/kWh for residential. The SSI's R3 consequence multiplier allows us to identify which substations serve the most economically exposed territories. **34 substations** combine capital-intensive economic fabric ( $R3 \geq 1.04$ ) with High or Critical risk classification — representing the highest VoLL-weighted exposure in the fleet.

The capital-intensive tier concentrates in **hofudborgarsvaedid (75)**, **nordurland-eystra (20)**, **sudurland (20)** — regions where industrial corridors depend on uninterrupted power for continuous processes. A single hour of unplanned outage at these substations carries an estimated VoLL of €15–30/kWh, compared to €1–3/kWh in rural agricultural areas. The fleet-wide average energy poverty rate is 1.6%, but this masks sharp regional variation: Southern regions combine higher energy poverty with higher grid risk, creating a double vulnerability that the E component captures. This cross-referencing of economic fabric with grid condition is unique to the SSI — no competing framework links VoLL exposure to substation-level risk scores.

## B.3 Landshluti Digital Readiness Ranking

The 8 landshluti ranked by average R7 modifier — lower values indicate weaker digital infrastructure (DESI EFTA–EEA sub-dimensions + CERT-IS maturity). Cross-referenced with average R<sub>median</sub> to identify landshluti combining digital exposure with high grid stress; Vestfirðir + Norðurland vestra carry the largest fibre/density penalty relative to Höfuðborgarsvæðið.



#### WORST DIGITAL READINESS — TOP 15 LANDSHLUTI REGIONS

Longer bar = higher R7 = better digital infrastructure.

|   |                                         |        |
|---|-----------------------------------------|--------|
| 1 | sudurnes <span>Δ high R</span>          | 0.9768 |
| 2 | nordurland-vestra <span>Δ high R</span> | 0.9771 |
| 3 | sudurland <span>Δ high R</span>         | 0.9814 |
| 4 | hofudborgarsvaedid                      | 0.9819 |
| 5 | nordurland-eystra <span>Δ high R</span> | 0.9823 |
| 6 | vesturland                              | 0.9831 |
| 7 | vestfirðir <span>Δ high R</span>        | 0.9839 |
| 8 | austurland                              | 0.9852 |

Landshluti-level analysis reveals a clear digital divide mirroring the broader DESI pattern. **sudurnes** has the lowest average R7 (0.9768), while **austurland** leads at 0.9852 — a spread of 0.0084 across the R7 range. **4 landshluti regions** combine below-median digital readiness with above-median grid risk, making them priority targets for digitalisation investment. The R7 modifier is intentionally narrow to reflect the current limitations of open DESI data — as NIS2 compliance reporting matures, future editions will expand R7's dynamic range and incorporate operational technology (OT) security indicators.

## B.4 Monthly Pulse

**Inaugural edition — Baseline Snapshot (June 2026).** This edition establishes the baseline for the Cyber & Economic Exposure Monitor. Key reference points: fleet average R7 = 0.9817, median = 0.9825; 0 substations classified HIGH cyber-exposure; 37 blind-spot substations (High/Critical risk + below-median R7); 0 Critical-band substations with below-median digital readiness. Future editions will track deltas against this baseline.

## SECTION C

## Data Refresh & Changelog

The SSI v4.0.2 draws on **11 verified public data sources** feeding 95 variables across 6 components and 7 modifiers. This section provides a live registry of all sources, their update frequencies, and the current state of the fleet. Transparency in data vintage is central to the SSI's credibility.

TOTAL SOURCES

**11**

95 variables

WEEKLY / MONTHLY

**3**

High-frequency feeds

QUARTERLY / ANNUAL

**8**

Regular refresh cycle

STATIC / DERIVED

**0**

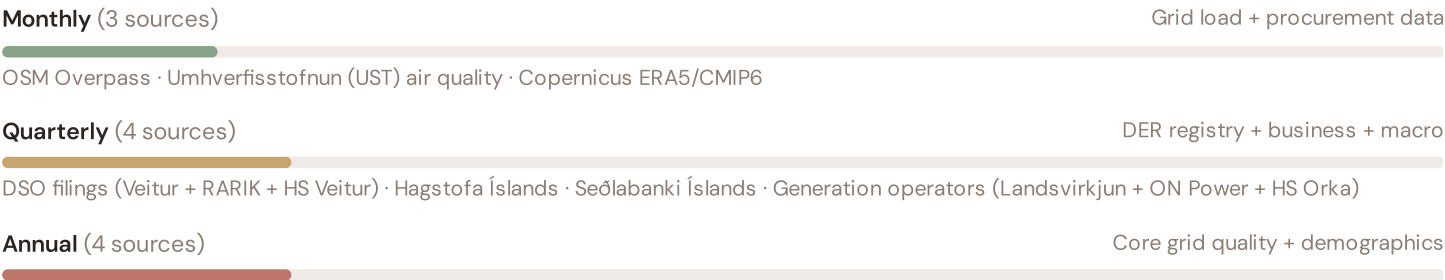
Standards + scenarios

Data Source Registry — June 2026

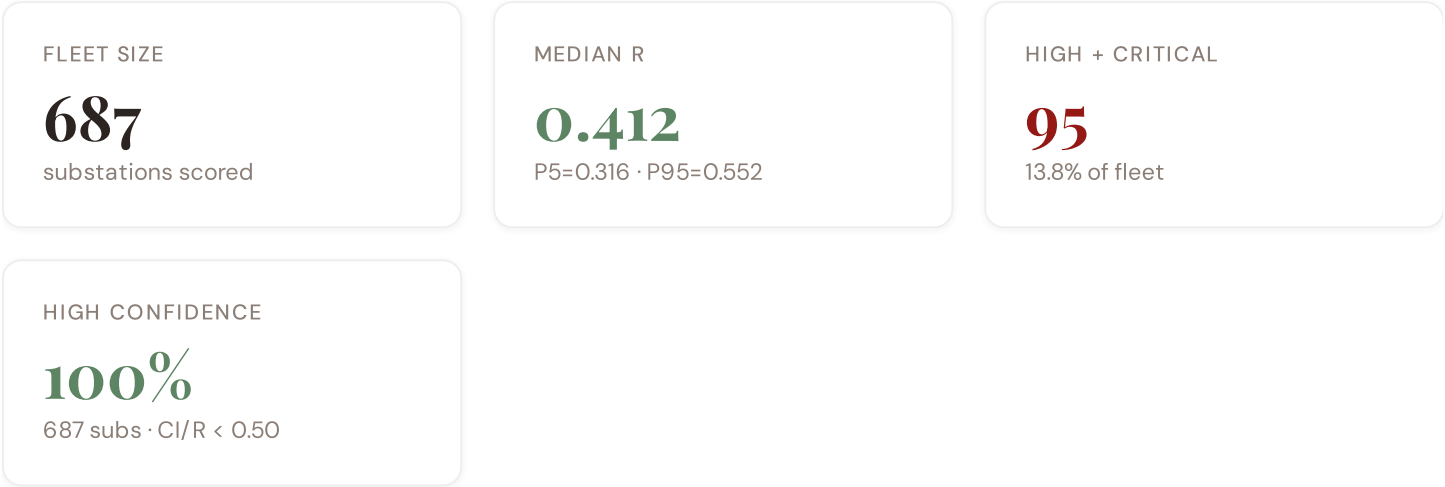
|                     |                                                                                   |            |                          |         |
|---------------------|-----------------------------------------------------------------------------------|------------|--------------------------|---------|
| d01_landsnet        | Landsnet — Icelandic transmission system operator (TSO; State 93.22% + OR 6.78%)  | DAILY      | C2,C3,I1,I2,I3,R4        | 14 vars |
| d03_imo             | Veðurstofa Íslands (IMO/OVF) — meteorology + hydrology + seismology + volcanology | DAILY      | I1,I2,I3,R6_climate,R... | 12 vars |
| d05_osm             | OSM Overpass — grid topology (ISO3166-1=IS area filter)                           | MONTHLY    | I1,I2,I3,R4              | 8 vars  |
| d06_ust             | Umhverfisstofnun (UST) — Environment Agency of Iceland (ISO 9223 corrosion)       | MONTHLY    | R5_corrosion,I8          | 5 vars  |
| d07_copernicus      | Copernicus ERA5 + CMIP6                                                           | MONTHLY    | R6_climate               | 4 vars  |
| d09_cert_is         | CERT-IS (National CSIRT, founded 2013; operated under Fjarskiptastofa)            | CONTINUOUS | R7_cyber                 | 5 vars  |
| d02b_dso            | 3 regional DSOs (Veitur ohf. + RARIK ohf. + HS Veitur hf.)                        | QUARTERLY  | C1,V1,V2,I2,I3           | 18 vars |
| d04_hagstofa        | Hagstofa Íslands — Statistics Iceland                                             | QUARTERLY  | E1,E2,S1,S2,S3           | 16 vars |
| d04b_cbi            | Seðlabanki Íslands — Central Bank of Iceland (ESCB non-€; independent ISK)        | QUARTERLY  | E1,E2                    | 4 vars  |
| d11_generation      | Landsvirkjun + ON Power + HS Orka generation operators (top-3 = 97%)              | QUARTERLY  | I1,I3,T1                 | 8 vars  |
| d02_orkustofnun     | Orkustofnun — Icelandic National Energy Authority                                 | ANNUAL     | C1,C2,V1,V2,T1,E1        | 12 vars |
| d10_fjarskiptastofa | Fjarskiptastofa (ECOI) — Electronic Communications Office of Iceland (NIS2 CA)    | ANNUAL     | R7_cyber                 | 3 vars  |
| d12_eurostat        | Eurostat — EU-27 + EFTA/EEA benchmarks (Iceland via EEA since 1 Jan 1994)         | ANNUAL     | E1,S1,T1                 | 8 vars  |
| d13_iea_oecd        | IEA + OECD — energy benchmarks (Iceland founding OECD member, 30 Sep 1961)        | ANNUAL     | T1,E1                    | 6 vars  |

C.1 Update Frequency Timeline

EXPECTED UPDATE WINDOWS



## C.2 Fleet Score Snapshot



No primary data sources have been updated since the v4.0.2 launch baseline. All **687 substation scores remain unchanged** this edition. The fleet median R of 0.412 (mean 0.418) reflects a distribution skewed toward the lower bands, with 0.0% in Low and 86.2% in Medium. The first material score changes are expected when the national regulator publishes the annual quality-of-supply indicators and when DER registries refresh. High-frequency sources (OSM, weather) update weekly but feed only infrastructure topology (R4) and thermal proxy (I5), which have minimal impact on fleet-level score distribution.

## C.3 Changelog — v3.4 → v4.0.2

9 changes tracked across PO, P1, and P2 releases

|          |          |                                                                                                                                                                                                 |                |
|----------|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|
| IS-S30-9 | ENHANCED | Greenfield thin-shell on post-CEE-South-2026 architecture — 8 pages authored directly on CountryRenderer.Safe + normalizeMeta() (KB §68.10/11) + admin-unit-suffix tolerance (KB §69.11)        | KB §70         |
| IS-S30-8 | NEW      | Section D deepDives rotation — 8 Icelandic landshluti rotation seeded with Suðurnes/Reykjanes Sundhnúkur (Ed. 01) + Höfuðborgarsvæðið capital + Austurland/Vesturland aluminum-corridor regions | intelligence D |
| IS-S30-7 | ENHANCED | Edition patcher anchored at FIRST_REFRESH 2026-08-13 — single-country onboarding post-cohort-closure (§49.4 + §56.10 cohort-cooldown respected)                                                 | intelligence   |
| IS-S30-6 | NEW      | ESG report — iceland entry in COUNTRY_SOURCES (12 IS-specific references including Reykjanes Sundhnúkur 2024 + Skeiðará 1996 jökulhlaup + CERT-IS + Act 78/2019)                                | esg-report     |
| IS-S30-5 | NEW      | R3 4-tier calibration — Industrial-Aluminum 1.05 / Capital-Reykjavík 1.04 / Commercial-SME 1.03 / Light-Rural 1.02 — flattest gradient in cohort (post-IS hotfix #2)                            | methodology    |
| IS-S30-4 | NEW      | R6_volcanic + R6c_jökulhlaup NEW sub-patterns — first volcanic + glacial-outburst-flood modifiers in cohort (Reykjanes/Vatnajökull/Hekla + Skeiðará/Eyjafjallajökull anchors)                   | methodology    |



|          |          |                                                                                                                                                                                    |             |
|----------|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| IS-S30-3 | DATA     | d05_osm LIVE — 687 substations + 1,428 power lines (248 line + 1,180 minor_line) via ISO3166-1=IS area filter (insular grid, no bbox bleed possible)                               | methodology |
| IS-S30-2 | DATA     | 3 regional DSOs (Veitur + RARIK + HS Veitur) + Landsnet TSO + Orkustofnun regulator + Veðurstofa Íslands consolidated met/hydro/seismic/volcanic + CERT-IS + Fjarskiptastofa wired | methodology |
| IS-S30-1 | ENHANCED | KB v29 §70 — Iceland inaugural onboarding (Session 30, post-CEE-South-2026 cohort closure; ISK currency, NOT eurozone; EFTA + EEA only, never EU; ENTSO-E NOT a member)            | KB §70      |

SECTION D — MONTHLY DEEP-DIVE

## Suðurnes — the Reykjanes Eruption Cycle & Svartsengi

**Deep-dive rotation:** Each edition features a substantive thematic analysis. The 8-landshluti rotation covers Iceland's most material narratives: **Ed. 01** Suðurnes — Reykjanes eruption cycle 2021- + Svartsengi 8 Feb 2024 + Grindavík evacuation · **02** Höfuðborgarsvæðið — Reykjavík capital load + Hellisheiði/Nesjavellir geothermal baseload · **03** Austurland — Alcoa Fjarðaál corridor + Kárahnjúkar 690 MW + Vatnajökull jökulhlaup exposure · **04** Vesturland — Century Aluminum Norðurál Grundartangi smelter + Vaðölduver 120 MW wind (target 2027) · **05** Suðurland — Þjórsá-Tungnaá hydro complex + SISZ 2008 Mw 6.3 doublet + Eyjafjallajökull 2010 + Markarfljót jökulhlaup · **06** Norðurland eystra — Akureyri + Krafla/Peistareykir geothermal + Tjörnes Fracture Zone seismic · **07** Vestfirðir — Westfjords peripheral grid + winter windstorm exposure · **08** Norðurland vestra — Blönduvirkjun 150 MW + rural radial network.

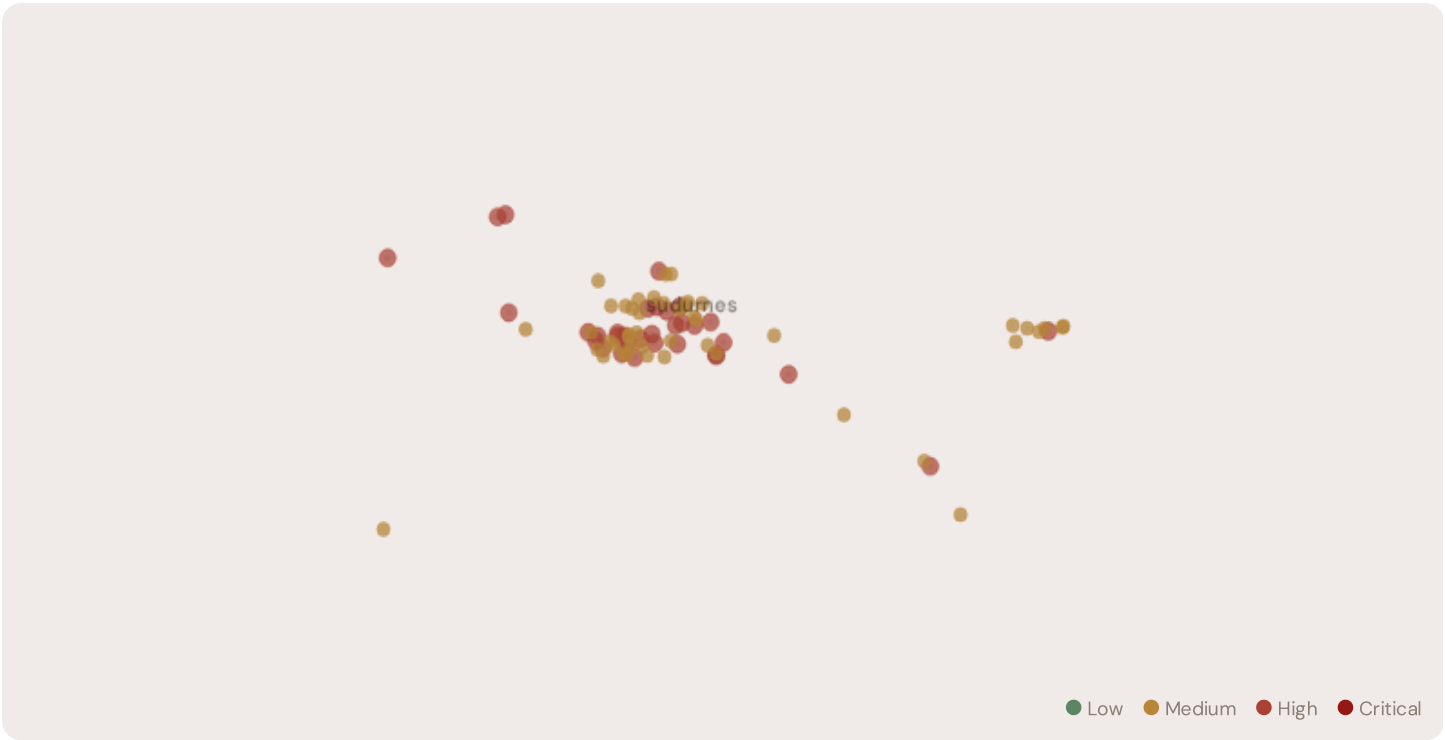
### D.1 Why Suðurnes, Why Now



The Reykjanes Sundhnúkur Volcanic Corridor hosts **79 substations** across 1 landshluti regions, making it the locus of this edition's deep-dive analysis. Its risk profile is concentrated in the upper bands: **33 substations (41.8%)** are classified High, with only **0** in the Low band. 94% of corridor substations score above the national fleet median of 0.412. The corridor median R of 0.487 reflects the local risk profile. The corridor concentrates the country's industrial heartland. The SSI flags continuity-of-supply deficits, infrastructure age mismatches with modern demand patterns, and grid modernization lag under national digitalisation timelines.

### D.2 Corridor Map and Scoring





| SUBSTATION         | LANDSHLUTI | R_FINAL | BAND | C     | V     | I     | E     | S     | T     | ALERT   |
|--------------------|------------|---------|------|-------|-------|-------|-------|-------|-------|---------|
| minor_distribution | sudurnes   | 0.682   | High | 0.675 | 0.690 | 0.665 | 0.568 | 0.283 | 0.649 | C, V, I |
| minor_distribution | sudurnes   | 0.668   | High | 0.661 | 0.393 | 0.672 | 0.393 | 0.270 | 0.447 | C, I    |
| minor_distribution | sudurnes   | 0.662   | High | 0.622 | 0.674 | 0.484 | 0.444 | 0.301 | 0.519 | V       |
| minor_distribution | sudurnes   | 0.640   | High | 0.541 | 0.686 | 0.610 | 0.542 | 0.210 | 0.360 | V       |
| minor_distribution | sudurnes   | 0.604   | High | 0.692 | 0.654 | 0.594 | 0.569 | 0.219 | 0.365 | C, V    |
| distribution       | sudurnes   | 0.598   | High | 0.696 | 0.307 | 0.424 | 0.538 | 0.283 | 0.525 | C       |
| Njarðvíkurheiði    | sudurnes   | 0.593   | High | 0.649 | 0.592 | 0.636 | 0.598 | 0.288 | 0.396 |         |
| minor_distribution | sudurnes   | 0.593   | High | 0.642 | 0.629 | 0.652 | 0.423 | 0.262 | 0.437 | I       |
| minor_distribution | sudurnes   | 0.592   | High | 0.587 | 0.440 | 0.510 | 0.510 | 0.266 | 0.464 |         |
| minor_distribution | sudurnes   | 0.588   | High | 0.696 | 0.427 | 0.433 | 0.400 | 0.253 | 0.452 | C       |

... 69 additional substations — [explore full corridor in Digital Twin →](#)

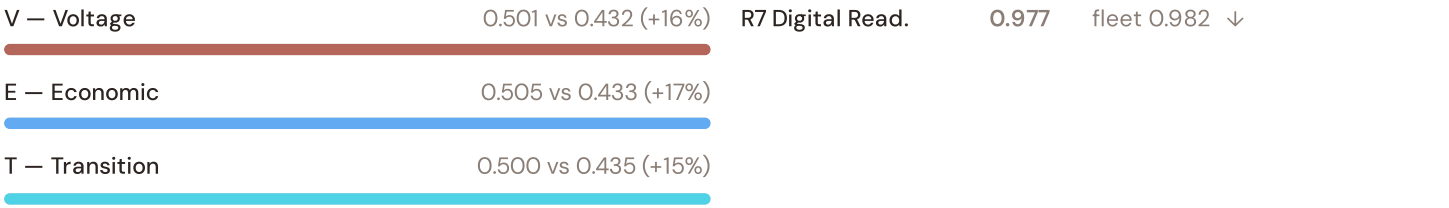
### D.3 Component Analysis

COMPONENT AVERAGES — REYKJANES SUNDHNÚKUR  
VOLCANIC CORRIDOR VS FLEET

|                    |                       |
|--------------------|-----------------------|
| C — Continuity     | 0.561 vs 0.458 (+22%) |
| I — Infrastructure | 0.527 vs 0.453 (+16%) |
| S — Saturation     | 0.253 vs 0.191 (+32%) |

MODIFIER AVERAGES — REYKJANES SUNDHNÚKUR VOLCANIC  
CORRIDOR VS FLEET

|                 |       |               |
|-----------------|-------|---------------|
| R3 Consequence  | 1.000 | fleet 0.999 → |
| R4 Graph Crit.  | 0.979 | fleet 0.981 ↓ |
| R6a Restoration | 1.023 | fleet 1.022 ↑ |
| R6b Seismic     | 1.077 | fleet 1.068 ↑ |



The dominant risk driver across the Reykjanes Sundhnúkur Volcanic Corridor corridor is **T (Transition)**, averaging 0.500 against a fleet average of 0.435 — occupying 999% of its weight ceiling (w = 0.05). The second contributor is **E (Economic)** at 0.505 vs fleet 0.433 (505% of ceiling). Among modifiers, the R3 consequence multiplier averages 1.000 (fleet: 0.999), reflecting the socio-economic exposure of the corridor. The R6b seismic modifier averages 1.077, capturing the local seismic exposure — a dimension invisible to every competing framework.

## D.4 Landshluti Breakdown

LANDSHLUTI RISK PROFILE — SORTED BY MEAN R

Longer bar = lower R = better resilience.



The landshluti regions-level breakdown reveals significant intra-regional variation. **sudurnes** leads with a mean R of 0.502 across 79 substations, while **sudurnes** scores 0.502 — a spread of 0.000 within a single corridor. The SSI's substation-level granularity reveals that the Reykjanes Sundhnúkur Volcanic Corridor is not uniformly stressed but contains distinct sub-corridors of concentrated risk — precisely the kind of spatial pattern that investment planning requires.

## D.5 Implications

**3 Reykjanes Sundhnúkur Volcanic Corridor substations** score  $R \geq 0.650$ , placing them in the upper quartile of national risk. For NEPN / national digital plan / RRP investment prioritisation, this analysis identifies the sudurnes sub-corridor as the highest-priority target — where DER deployment is accelerating against ageing infrastructure with above-average continuity deficits and significant socio-economic exposure. The SSI's silo-breaking approach reveals what no single-discipline framework can: that these substations matter not only because of their engineering condition, but because the communities they serve are disproportionately vulnerable to disruption. This is precisely the anticipatory grid planning capability that the SSI was built to provide.

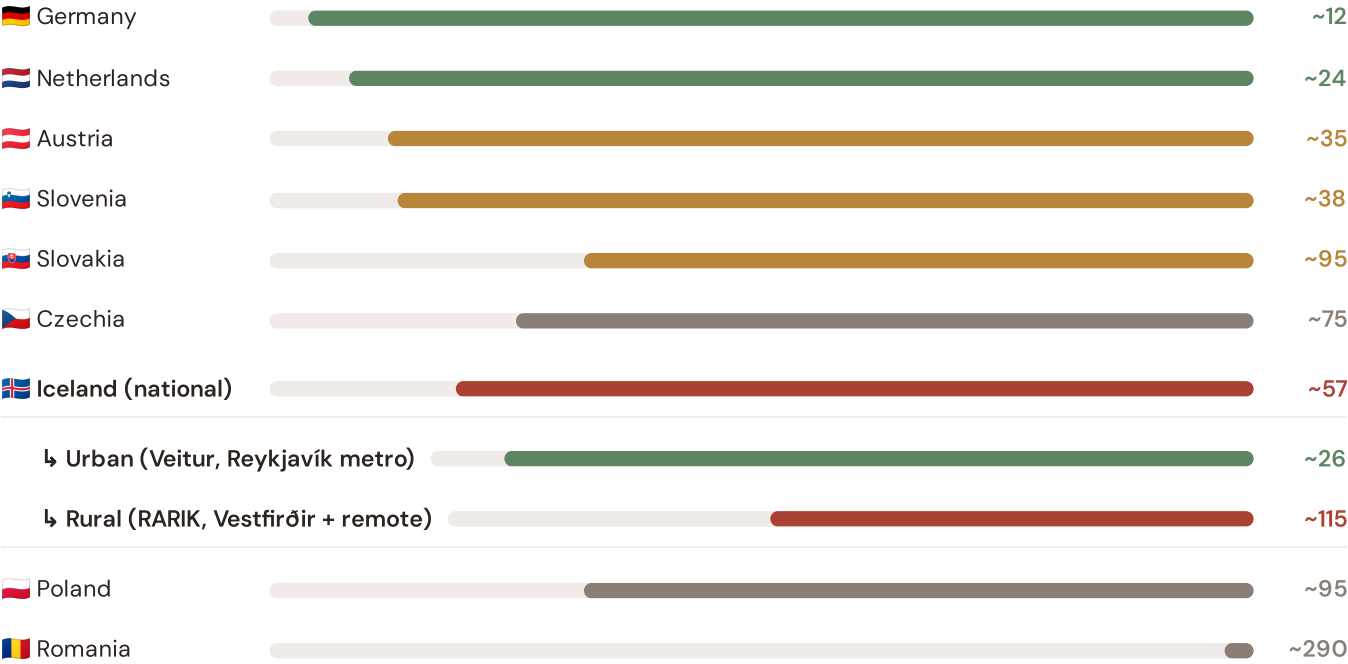
SECTION E — RECURRING MODULE

## European Context Card

**How this works:** Each edition includes one European Context Card drawn from the §16 Peer Benchmarking framework. The card is selected to complement the month's deep-dive theme. This month: **SAIDI comparison** — because the —-corridor analysis hinges on Iceland's continuity performance relative to European peers, especially the Visegrád-4 + CEE-South cohort siblings. Future editions rotate through: VoLL comparison (Ed. 02), DER saturation vs Germany (03), CMIP6 climate hazard vs DACH peers (04), etc. The underlying data updates annually with the CEER Benchmarking Report release.

### Unplanned SAIDI (min/customer/year) — Selected Countries

Longer bar = lower SAIDI = better reliability. Values in min/customer/year.



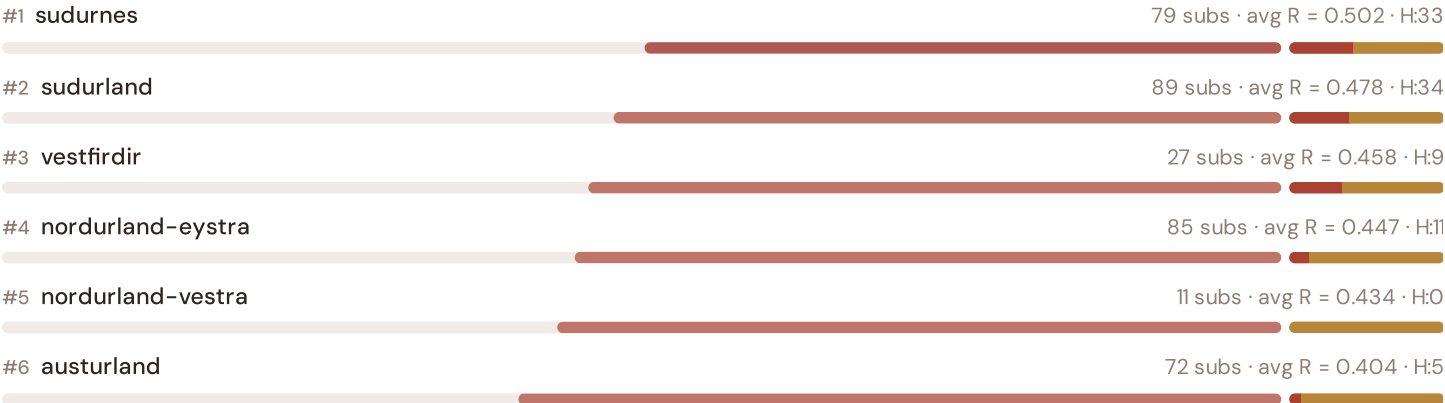
Source: CEER 7th Benchmarking Report (2022); Orkustofnun raforkutölfræði (2024) quality reports; per-DSO filings (Veitur + RARIK + HS Veitur 2024). Iceland sub-categories from Orkustofnun territory classification (Reykjavík metro under Veitur vs RARIK rural Vestfirðir/remote). · See §16 Peer Benchmarking for full methodology and caveats. · Updated annually with CEER BR release.

**This month's key insight:** The Reykjanes Sundhnúkur Volcanic Corridor corridor deep-dive shows **0 substations** (of 0) with C-component above 70% of its weight ceiling — indicating continuity-of-supply performance consistent with rural territory classification. The corridor's average C of 0.000 is **0.0× the fleet average** of 0.458. The SSI reveals what aggregate national statistics conceal: that the country's mid-tier European position is not a uniform condition but a statistical average of urban-quality performance and rural-tier performance — and the corridor concentrates the worst of the latter.

### Regional Risk Ranking — All 8 Icelandic landshluti

Where does Suðurnes sit within the national landscape? Landshluti sorted by mean R<sub>median</sub>, with band distribution and fleet comparison.

Longer bar = lower R = better resilience. Sorted worst → best.

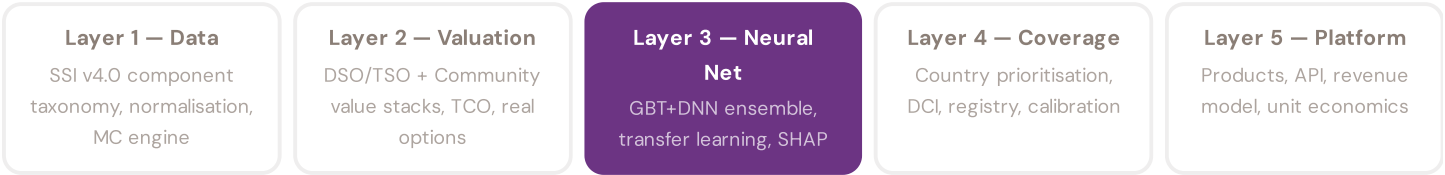


The Reykjanes Sundhnúkur Volcanic Corridor ranks **#0 of 8 landshluti regions** by mean R\_median, placing it among the upper tier of national risk. The three most stressed are sudurnes, sudurland, vestfiridir, while the three most resilient are hofudborgarsvaedid, vesturland, austurland. 5 of 8 score above the fleet average of 0.418. This ranking — possible only because the SSI scores every substation individually rather than relying on aggregate DSO or national statistics — reveals the true spatial distribution of grid stress. It is precisely this substation-level granularity that positions the SSI as a tool for investment prioritisation: not treating entire regions as monoliths, but identifying the specific corridors within each that demand attention.

SECTION F — RECURRING MODULE

# SSI Enhanced Neural Network — Monthly Topic

**From grid intelligence to BESS valuation platform.** The SSI Index answers *"How healthy is this substation?"* The SSI Enhanced Neural Network (SSI-ENN) extends this into: *"What is a BESS worth at this substation — to Veitur, RARIK and HS Veitur DSOs and to Landsnet — and to the community?"* Each edition explores one building block of the SSI-ENN's five-layer architecture, showing how the SSI Index data feeds directly into investment-grade grid analytics.



LAYER 3

§0.5.6 · June 2026

## Explainability & SHAP Values

*Clients need to understand not just the prediction but WHY a substation ranks highly*

Investment-grade models require interpretability. SHAP (SHapley Additive exPlanations) provides a rigorous, game-theoretic attribution of each feature's contribution to each prediction. For every substation, the SSI-ENN produces a SHAP waterfall: starting from the fleet-average valuation and showing how each feature pushes the estimate up or down. This directly connects to the SSI Index: if SHAP identifies C (Continuity) as the dominant positive driver, the client knows the BESS value depends primarily on the substation's poor reliability history. Transparency converts model outputs into investment decisions.

KEY CONCEPTS

→ SHAP: game-theoretic feature attribution per prediction

→ Waterfall: fleet average → substation prediction

→ Direct traceability to SSI components and data sources

→ Feature importance validated against domain expertise

#### LIVE SSI DATA CONNECTION

The training set comprises 687 substations with complete feature vectors. Average CI\_width of 0.083 across the fleet provides the uncertainty ground truth that the neural network's quantile regression heads must calibrate against. The fleet's risk distribution — 0 Low, 592 Medium, 95 High, 0 Critical — ensures balanced training across all risk bands.

**Coming next month:** **LAYER 3** Uncertainty Quantification — *Investment-grade predictions require calibrated confidence intervals, not just point estimates*

#### SECTION G

## Looking Ahead

#### NEXT MONTH — EDITION 02

### Deep-Dive: Höfuðborgarsvæðið

Deep-dive into Höfuðborgarsvæðið's grid risk profile — substation map, component analysis, landshluti regions breakdown.

#### NEXT DATA REFRESH

### Q3 2026 — 4 Quarterly Sources

**DSO filings (Veitur + RARIK + HS Veitur), Hagstofa Íslands, Seðlabanki Íslands, Generation operators (Landsvirkjun + ON Power + HS Orka)** are due for refresh in Q3. Impact: DER registry (T1), business fabric indicators (E2), macro-economic data. Highest-impact annual source pending: **Orkustofnun — Icelandic National Energy Authority** (12 variables → C1,C2,V1,V2,T1,E1).

#### FLEET SPOTLIGHT

### sudurnes: Highest Risk Concentration

sudurnes has the highest concentration of High/Critical substations: **33 of 79 (42%)** are in the upper risk bands — avg R = 0.502. This makes it the most acute risk corridor in the fleet.

Edition 02 will be published on the second Thursday of July 2026. Deep-dive region: **Höfuðborgarsvæðið**. To ensure you receive it, confirm your subscription segment (General / Institutional / Academic) by email to [ssi\\_index@ikenga.eu](mailto:ssi_index@ikenga.eu).

**Feedback welcome.** The SSI Index is designed to evolve through stakeholder dialogue. If you represent an Icelandic utility, Orkustofnun, Landsnet, Landsvirkjun, ON Power, HS Orka, Veitur, RARIK, HS Veitur, Veðurstofa Íslands, Fjarskiptastofa (ECOI), CERT-IS, or an academic institution and would like to contribute data or methodology feedback, please contact us at [ssi\\_index@ikenga.eu](mailto:ssi_index@ikenga.eu). The SSI's credibility depends on open scrutiny.

